

KHALLAF TAHER

Cloud & IoT Engineering Student · HENNGE GIP Candidate

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Engineering Talent Acquisition Team

HENNGE K.K. · Daiwa Shibuya Square, Nanpeidaicho, Shibuya-ku, Tokyo, Japan

Re: Application for the Full Stack Engineering Pathway — HENNGE Global Internship Program (GIP)

Dear HENNGE Global Internship Team,

I am writing to express my strong enthusiasm for applying to the Full Stack Engineering pathway of the HENNGE Global Internship Program (GIP). What drew me to HENNGE is how authentically your team embodies an open engineering culture. Most tech companies talk about agility and transparency, but reading HENNGE's engineering blog — from honest technical post-mortems and hiring philosophies to your active contributions to Open Source Software (OSS) — revealed an organization that truly practices what it preaches. That culture of openness, paired with your leadership as Japan's premier cloud security and SaaS innovator, is precisely where I want to challenge myself and contribute.

At HENNGE, you define a standout engineer as a "*learnaholic*" who embraces **Henka (transformation) and continuous challenge**. This philosophy has defined my entire engineering journey. Coming from the rigorous mathematical and analytical background of CPGE preparatory classes, I taught myself modern full-stack web development and cloud infrastructure from the ground up, driven by a relentless curiosity to understand how distributed software operates from kernel to client.

The project where this clicked into place is my **Cloud Notes API** — a containerized application built with FastAPI, PostgreSQL, and Docker, deployed live on AWS EC2 with automated CI/CD via GitHub Actions and Terraform IaC. Architecting the backend with stateless JWT authentication and `slowapi` rate limiting was deeply satisfying, but my most exciting breakthrough came when building the React frontend interface. Transitioning from a server-side mindset to managing client state and reactive UI components was unfamiliar at first — until I created a note and watched it render instantaneously without a full-page reload. That excitement of bridging resilient backend systems with fluid, human-centric user experiences is what defines full-stack engineering for me. I have since extended this platform with an AI Retrieval-Augmented Generation (RAG) vector layer using Claude 3.5 Sonnet to enable natural language querying across notes.

I continued pushing this boundary in my recent **3D IoT Digital Twin Platform**. Combining an AWS IoT Core MQTT bridge over TLS (X.509 certs), cloud persistence on AWS RDS PostgreSQL 16, and bi-directional actuation APIs with an interactive 3D WebGL dashboard built in Three.js and real-time WebSockets showed me the power of transforming raw telemetry into intuitive, actionable visualizations.

Beyond code, I believe that continuous learning is magnified when shared with others. As **Head of the English Debating Cell at INPT**, I noticed many brilliant engineering peers struggled with expressing their technical ideas. I designed and led a 14-session onboarding program to mentor students through British Parliamentary debate formats, helping them build structured reasoning and overcome public speaking anxiety. Mentoring juniors and fostering an inclusive, communicative environment taught me that great engineering teams succeed through empathy, clarity, and shared momentum.

My decision to pursue my career in Japan is intentional and proactive. I have been self-studying Japanese and have actively enrolled in **The University of Tokyo's Global Consumer Intelligence (UTokyo GCI)** program to deepen my engagement with Japanese data science and engineering methodologies. The HENNGE Global Internship is not a temporary stop for me; it is the dedicated, deliberate first step toward building a long-term engineering career in Tokyo. I would be thrilled to bring my energy, full-stack skillset, and learnaholic mindset to HENNGE.

Thank you for your time, consideration, and dedication to reviewing each application with human care.

With deep respect and enthusiasm,

KHALLAF Taher

Cloud & IoT Engineering, INPT Rabat — UTokyo GCI Enrolled